Cell Design Specification — combo-v5-final (t2_r3)

VBF Case t2_r3 | VBF-T2R3-DS-01 | Virtual simulation calibre | Generated 2026-08-26

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All values mechanically sourced from `cell/r7_final_*_dfn.json`, `cell/r6_combo-v5-final_aging*_spme.json`, `r6_combo-v5-final_params.json` and the Chen2020 parameter set; line-by-line source below. No values written from memory.

1. Basic specification

Field	Value	Source
Electrochemical system	NMC811 / graphite	Chen2020 base set (task names no system)
Nominal capacity	5.0 Ah (parameter set) / 5.0648 Ah (simulated)	`Nominal cell capacity [A.h]`; `r7_final_1c_dfn.json` capacity_ah
Voltage window	2.5–4.2 V	parameter set cut-offs
Cell dimensions (H×W×T stack)	65 mm × 1580 mm × 0.201 mm	`Electrode height/width [m]`; thickness = 75.6+12.0+85.2+16+12 µm layer sum (calc-energy thickness_m)
Shell thickness	Not provided (no parameter)	—
Electrolyte formulation	1 M $LiPF_6$ in EC:EMC (3:7 w/w) — Chen2020 parameterization; additive candidates: Not provided (not parameterized)	Chen2020 set documentation
Cation transference number	0.45 (override; baseline 0.2594)	`r6_combo-v5-final_params.json` / Chen2020
Electrolyte conductivity	1.7 S/m constant (override; baseline Nyman2008 σ(298 K)=0.9487 S/m)	params / Chen2020 function
Electrolyte diffusivity	5.0e-10 m²/s constant (override; baseline 1.769e-10)	params / Chen2020 function

2. Electrode and separator

Layer	Thickness	Porosity	Active vol. frac.	Particle radius	Material / collector
Positive (NMC811)	75.6 µm	0.335	0.665	2.5 µm (final; baseline 5.22)	ρ = 3262 kg/m³
Negative (graphite)	85.2 µm	0.4 (final; baseline 0.25)	0.75	2.0 µm (final; baseline 5.86)	ρ = 1657 kg/m³
Separator	12 µm	0.47	—	—	ρ = 397 kg/m³
Positive collector	16 µm Al	—	—	—	ρ = 2700 kg/m³
Negative collector	12 µm Cu	—	—	—	ρ = 8960 kg/m³

All parameters above are Chen2020 set values except porosities/particle radii overridden by `r6_combo-v5-final_params.json`.

N/P ratio = 0.667 (mechanical: c_max×active-volume-fraction×thickness, neg/pos = 33133×0.75×85.2 / 63104×0.665×75.6 µm). Below unity means the parameter set's positive electrode is oversized relative to the negative (anode-limited window). Set-inherent; unchanged by this design's overrides; flagged for physical cell-balancing validation. µm unit note: ratio is monotonic in thickness so units cancel.

3. Process design parameters

Parameter	Value	Formula	Source
Pos. areal density	163.99 g/m²	L×(1−porosity)×ρ	75.6 µm×0.665×3262 kg/m³ (Chen2020)
Neg. areal density	84.71 g/m²	L×(1−porosity)×ρ	85.2 µm×0.6×1657 kg/m³
Pos. compaction density	2.17 g/cm³	ρ×(1−porosity)÷1000	divided by 1000 (kg/m³→g/cm³)
Neg. compaction density	0.99 g/cm³	ρ×(1−porosity)÷1000	
Electrolyte fill amount	8.02 g	pore volume×1.2 g/cm³×fill factor 1.0	pore = 6.7 cm³; electrolyte density 1.2 g/cm³ literature value (not in set)
Formation recommendation	0.1C CC to 4.2 V, 25 °C, 2 cycles	—	design recommended value; actual production-line value requires tuning, flagged for validation

4. Mass breakdown

Component	Mass	Formula-calibre note
Positive electrode (NMC811 × 0.665 vol)	16.8422 g	L×A×(1−por)×ρ from calc-energy layer_kg_m2 = 0.163993788 kg/m² × 0.1027 m²
Negative electrode (graphite × 0.75 vol)	8.6993 g	layer_kg_m2 = 0.08470583999999999 kg/m² × 0.1027 m²
Separator	0.2593 g	layer_kg_m2 = 0.00252492 kg/m² × 0.1027 m²

Positive collector Al	4.4366 g	layer_kg_m2 = 0.043199999999999995 kg/m^2 × 0.1027 m^2
Negative collector Cu	11.0423 g	layer_kg_m2 = 0.10752 kg/m^2 × 0.1027 m^2
Electrolyte (excluded by contract calibre)	8.02 g	pore-volume calibre, literature 1.2 g/cm^3
Total (calc-energy contract calibre, electrolyte excluded)	**41.2797 g**	`r7_final_energy_dfn.json` mass_kg = 0.04128 kg
Total incl. electrolyte	49.30 g	auxiliary calibre

5. Performance verification

Metric	Simulated	Threshold (entry 0)	Determination	Source
Energy density	447.12 Wh/kg	≥ 327.18	✓ PASS	`r7_final_energy_dfn.json` energy_density_wh_kg
Low-T retention (–20 °C/25 °C, DFN)	0.9957	≥ 0.90	✓ PASS	`r7_final_lowT_retention.json` = 5.0430/5.0648
SEI @ 100 cyc	276.2 nm	≤ 500	✓ PASS	`r7_final_sei_thickness_nm_100cyc.json` (SPMe aging, note in file)
SEI @ 500 cyc	393.0 nm	≤ 550	✓ PASS	`r7_final_sei_thickness_nm_500cyc.json`
4C plating	anode potential min = +0.0443 V	≥ 0 V (no plating)	✓ PASS	`r7_final_4c_dfn.json` anode_potential_v
4C fast charge	charge-phase duration ≈ 693 s (V-min restart → 4.2 V cut-off at 4344.8 s); charge capacity 0.7705 Ah; protocol = 1C pre-discharge (4.106→2.956 V, ~3530 s) + 4C CC charge, 318.15 K ambient	recorded	—	`r7_final_4c_dfn.json` time_s / voltage_v / capacity_ah (runner: C-rate 4.0)
Max cell temperature @ 4C	347.70 K (74.55 °C)	no contract threshold; recorded	—	4C protocol runs at 318.15 K ambient (`run-pyamm --thermal lumped`)
1C capacity	5.0648 Ah	nominal 5.0 Ah	recorded	`r7_final_1c_dfn.json`

6. Design notes — parameters changed vs Chen2020 baseline

Parameter	Baseline	Final	Why (citing evaluate-log rounds)
Positive particle radius	5.22 µm	2.5 µm	unlocks 4C charge acceptance: baseline cathode surface saturates ~26 s into 4C (t≈R^2/D_s hours); fine particles shorten diffusion (round 3 evidence; r4-r7)
Negative particle radius	5.86 µm	2.0 µm	raises anode charge-transfer area, lifts anode potential away from 0 V during 4C (rounds 2/5/7)
Electrolyte conductivity	σ(T) fcn (0.9487 @298K)	1.7 S/m const.	transport override; reduces ohmic drop at 4C and –20 °C; literature-class constant bridge flagged as estimate (rounds 2/6)
Electrolyte diffusivity	1.769e-10 m^2/s	5e-10	concentration-polarization relief; supports low-T retention (rounds 2/6)
Cation transference number	0.2594	0.45	further concentration-polarization relief (rounds 2/6)
Negative porosity	0.25	0.40	electrolyte path widening → anode potential lift during 4C; small ED cost (round 5/6)
SEI kinetic rate constant	1e-12 m/s	2e-13	SEI-suppressing-coating bridge; near-inert at 100 cyc (round 2) but retained for late-life (rounds 2–7)
SEI partial molar volume	9.585e-5 m^3/mol	4.7925e-5	denser, LiF/Li■O-rich SEI film: thickness scales linearly with V■, no early-life IR penalty (rounds 5/6 discovery)

Bridge/estimate flags: transport overrides are literature-class constants (not simulation-derived); SEI k/V■ are electrode-modification bridges; all flagged per protocol as fabrication-grade estimators, re-validatable by true stage 2–4 (skipped: real_compute=false).